www.tntextbooks.in

Differences Between Vascular Cambium and Cork Cambium

Vascular cambium	Cork cambium
Also called cambium	Also called phellogen
It arises from procambium and interfascicular parenchyma in stems and from conjunctive parenchyma in roots	It arises from epidermis, cortex, phloem, or pericyle in both stems and roots
It comprises long fusiform and short ray initials.	It comprises of homogenous cells.
It produces secondary phloem towards the outer side and secondary xylem towards inner side.	It produces phellem(cork) towards outer side and phelloderm (secondary cortex) towards inner side.

Bark

The term 'bark' is commonly applied to all the tissues outside the vascular cambium of stem (**i.e., periderm, cortex, primary phloem and secondary phloem**). Bark protects the plant from parasitic fungi and insects, prevents water loss by evaporation and guards against variations of external temperature. It is an insect repellent, decay proof, fireproof and is used in obtaining drugs or spices. The phloem cells of the bark are involved in conduction of food while secondary cortical cells involved in storage.

If the phellogen forms a complete cylinder around the stem, it gives rise to **ring barks**. Example: *Quercus*. When the bark is formed in overlapping scale like layers, it is known as **scale bark**. Example: Guava. While ring barks normally do not peeled off, scale barks peeled off.



Figure 10.14: *Quercus* Tree-showing ring bark



Figure 10.15: Guava tree showing scale bark

Lenticel

Lenticel is raised opening or pore on the epidermis or bark of stems and roots.

It is formed during secondary growth in stems. When phellogen is more active in the region of lenticels, a mass of loosely arranged thin-walled parenchyma cells are formed. It is called **complementary tissue** or **filling tissue**.

Lenticel is helpful in exchange of gases and transpiration called **lenticular transpiration**.